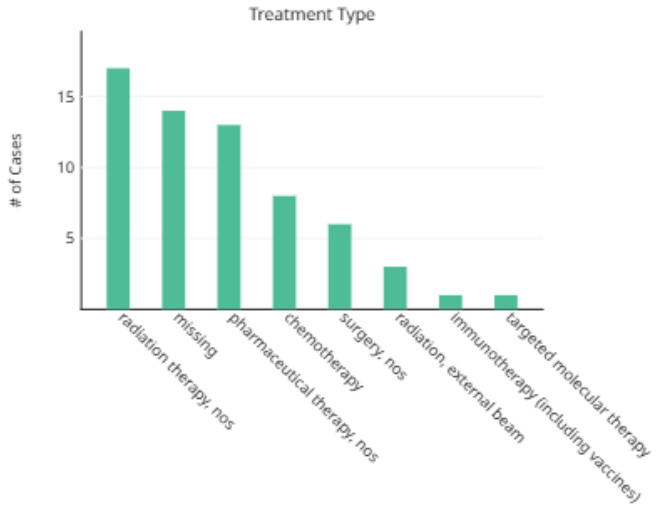
Characterizing Gender and Subtype-Specific Molecular Signatures in Non-Smoking NSCLC: A Multi-Omics Approach to Personalized Therapy

INTRODUCTION

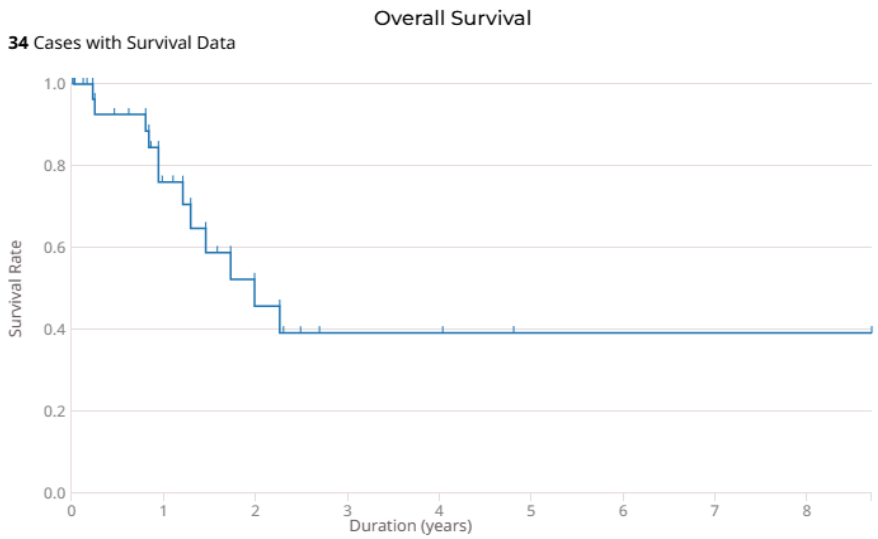
Lung cancer is the leading cause of cancer-related deaths worldwide.[1] While smoking is the primary risk factor, a significant number of cases occur in non-smokers. Non-smoking-related risk factors include occupational carcinogen exposure, radon, air pollution, certain infections, and prior chest radiation.[2] This study focuses on using a multi-omics approach—encompassing genomics, transcriptomics, epigenomics, and pharmacogenomics—to optimize personalized therapies for NSCLC, with particular attention to subtype-specific, gender-specific, and non-smoker-related variations. NSCLC accounts for approximately 85% of lung tumors, with lung adenocarcinoma (LUAD) and squamous cell carcinoma (LUSC) being the most common subtypes.[3],[4] Due to its prevalence and relatively slower progression compared to small-cell lung cancer (SCLC), NSCLC is an ideal target for the development of targeted therapies. However, targeted therapies are not yet widely implemented in clinical practice, primarily because the underlying causes of the disease are not yet well understood. There is an urgent necessity to integrate knowledge of all biological pathways to effectively treat cancer, as it is a heterogeneous disease and manifests differently in each patient.

LIMITED ACCESS TO TARGETED THERAPIES IN NON-SMOKING NSCLC PATIENTS

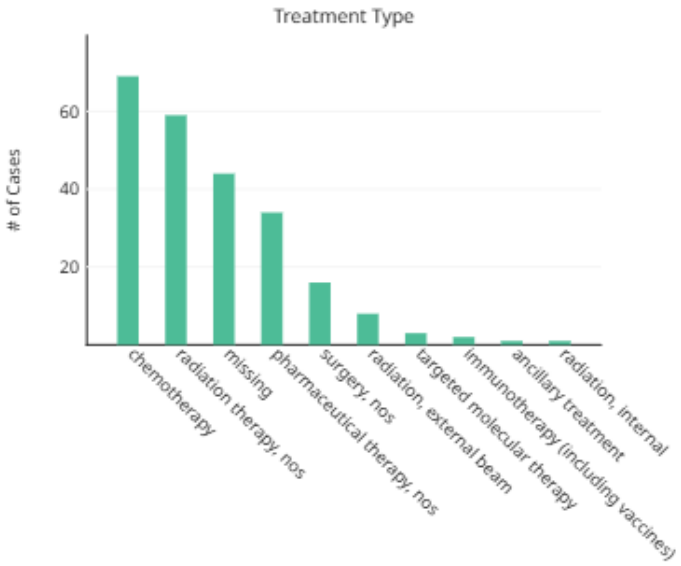
Among non-smoking patients with NSCLC, the use of targeted therapies is really low. In adenocarcinoma, only 3 out of 121 patients received targeted treatment (1.27%). Similarly, among patients with lung squamous carcinoma, only 1 received targeted therapy (1.59%). Survival curves show that patients with squamous cell carcinoma have a 39% survival rate, while those with adenocarcinoma have a 40% survival rate. Because most patients were treated primarily with radiation therapy—which does not address the underlying molecular drivers of disease—these low survival outcomes highlight a significant therapeutic gap. Increasing access to targeted therapies could improve survival by addressing the disease's molecular cause.



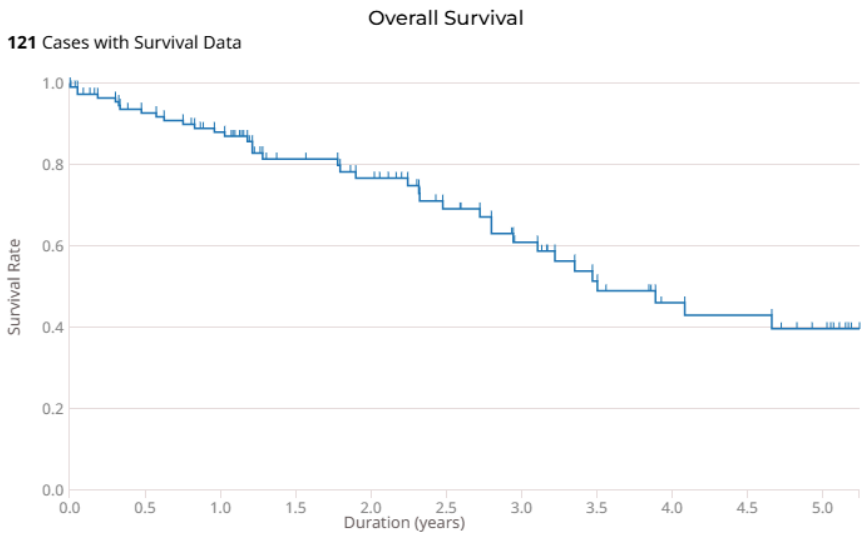
Treatments received by non-smoking patients with squamous cell carcinoma



Survival rate of non-smoking patients with squamous cell carcinoma



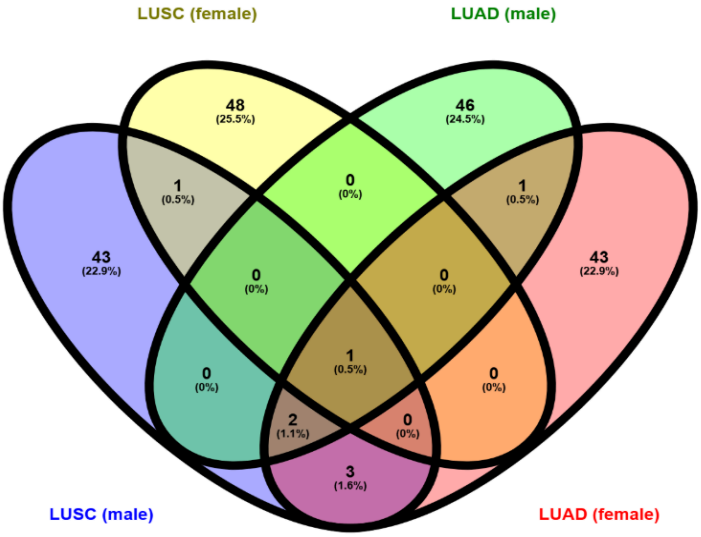
Treatments received by non-smoking patients with adenocarcinoma



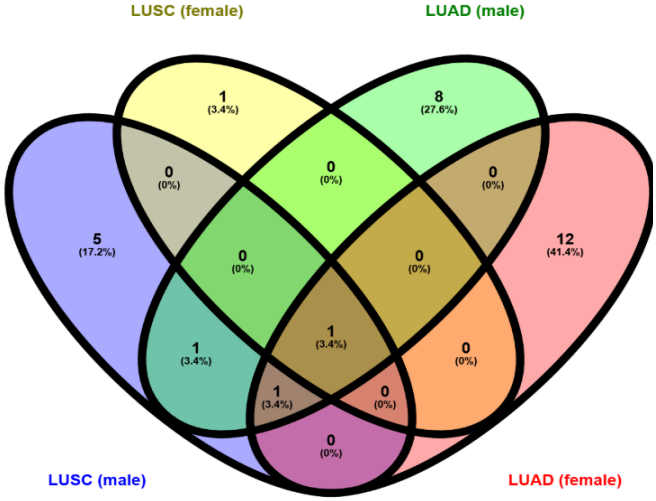
Survival rate of non-smoking patients with adenocarcinoma

MOLECULAR HETEROGENITY IN NSCLC SUBTYPES

Among non-smokers, stratified by lung cancer subtype (LUSC and LUAD) and gender, the TP53 gene is mutated across all four subgroups. TP53 is also one of the most commonly mutated genes in various cancers, as it plays a critical role in regulating programmed cell death (apoptosis). Mutations in TP53 promote prolonged cell proliferation and allow cells to evade apoptosis, contributing to tumor development. Another key driver gene is EGFR (Epidermal Growth Factor Receptor), which is associated explicitly with LUAD. Unlike in smokers, where EGFR mutations were observed only in females, in non-smokers, this gene is mutated in both males and females, suggesting that smokers and non-smokers may require different treatment strategies. KRAS mutations are more common in adenocarcinoma among non-smoking patients, indicating a subtype-specific association. In contrast, KEAP1 mutations occur more frequently in males across both LUSC and LUAD subtypes, suggesting male-specific prevalence. Additionally, TTN mutations appear to be specific to adenocarcinoma, whereas ZAN mutations are specific to squamous cell carcinoma, with amplifications observed in FGFR1 and SOX2.

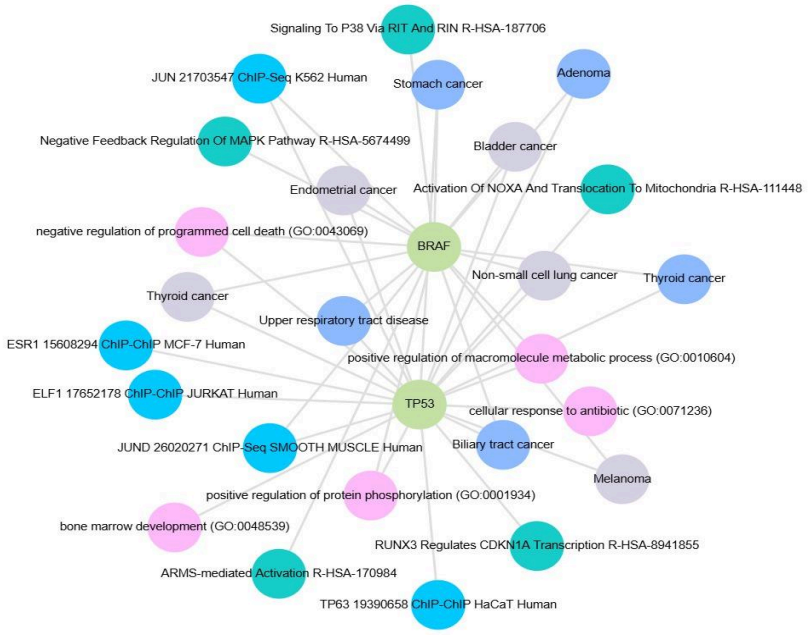


Venn Diagram 1: *Shows shared mutations among the four subgroups: LUSC male, LUSC female, LUAD male, and LUAD female.*



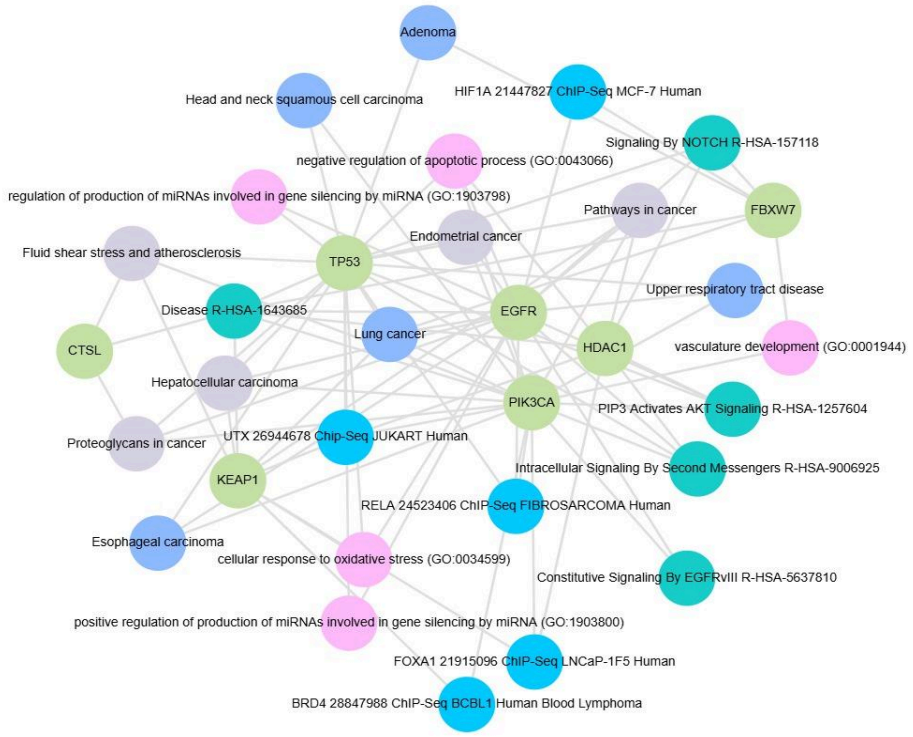
Venn Diagram 2:*Shows shared driver oncogenes among the four subgroups: LUSC male, LUSC female, LUAD male, and LUAD female.*

ENRICHMENT NETWORKS OF KEY PATHWAYS AND GENE INTERACTION IN NSCLC SUBTYPES STRATIFIED BY SEX



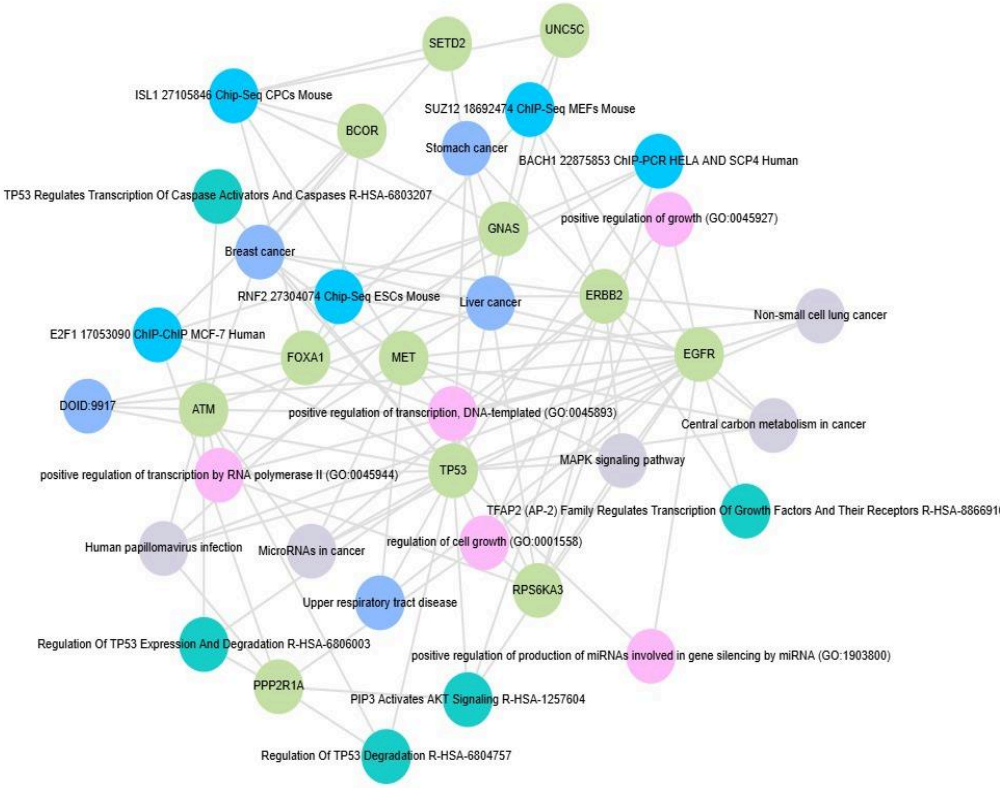
Enrichment network of driver genes in female lung squamous carcinoma

The enrichment network reveals that **BRAF** and **TP53** participate in the **negative regulation of programmed cell death** (GO:0043069), emphasizing their role in helping cancer cells evade apoptosis. **TP53** uniquely contributes to **bone marrow development** (GO:0048539), suggesting that mutations in this gene may lead to **anemia** and **immunodeficiency**, as bone marrow is critical for the production of blood and immune cells. Both genes are also associated with **upper respiratory tract disease**, indicating that pre-existing respiratory conditions could contribute to lung cancer progression. **Reactome pathway analysis** shows that TP53 is involved in **RUNX3-mediated transcription of CDKN1A** and **NOXA activation**, reinforcing its role in the **DNA damage response** and **regulation of apoptosis**. **BRAF**, in contrast, participates in **negative feedback regulation of the MAPK pathway**, **ARMS-mediated activation**, and **p38 signaling via RIT/RIN**, all of which are essential for **tumor growth** and **stress response**. **Transcription factor enrichment (ChEA3)** indicates that TP63, ESR1, and ELF1 regulate TP53, while both BRAF and TP53 are targets of **JUND** and **JUN**, reflecting shared upstream transcriptional control. Notably, TP53 in **female LUSC** is regulated by **ESR1 (estrogen receptor 1)**, suggesting potential **hormonal regulation** of this tumor suppressor gene.



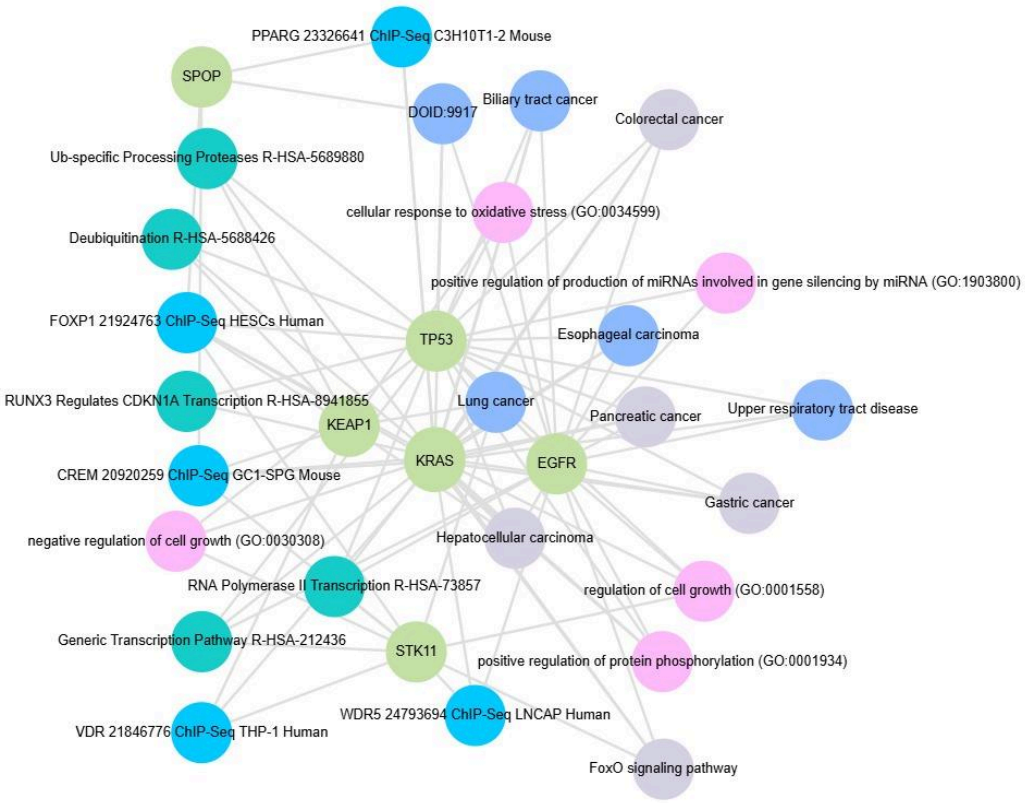
Enrichment network of driver genes in male lung squamous cell carcinoma

The enrichment network for male LUSC driver genes shows that **PIK3CA, EGFR, HDAC1, KEAP1, FBXW7, and TP53** are involved in the **negative regulation of apoptosis (GO:0043066)**, underscoring their role in promoting cancer cell survival. **EGFR and TP53** also contribute to the **positive regulation of miRNA production (GO:1903800; GO:1903798)**, linking them to miRNA-mediated gene silencing, a critical post-transcriptional regulatory mechanism. Additionally, **EGFR, KEAP1, and TP53** participate in the **cellular response to oxidative stress (GO:0034599)**, while **PIK3CA and FBXW7** are involved in **vasculature development (GO:0001944)**, highlighting their role in tumor angiogenesis. These genes are also associated with **upper respiratory tract disease**, suggesting that pre-existing respiratory conditions may indicate higher cancer susceptibility. According to the KEGG pathway analysis, these driver genes are implicated in **Fluid Shear Stress and Atherosclerosis**. Furthermore, the Reactome analysis shows that they participate in key signaling cascades, including **EGFRvIII constitutive signaling**, **NOTCH signaling**, **intracellular second messenger pathways**, and **PIP3-AKT activation**, all of which are central to tumor growth, survival, and therapeutic resistance. Transcription factor analysis from ChEA3 indicates that **RELA, HIF1A, FOXA1, BRD4, and UTX** regulate subsets of these genes, with **PIK3CA, EGFR, and TP53** under coordinated control by **RELA**.



Enrichment network of driver genes in female lung adenocarcinoma

The enrichment network for **female LUAD driver genes** reveals that **TP53 and EGFR** are involved in the **positive regulation of miRNA production for gene silencing** (GO:1903800), highlighting their role in post-transcriptional regulation. **EGFR, PPP2R1A, ERBB2, RPS6KA3, and TP53** participate in the **regulation of cell growth** (GO:0001558), while **MET, ATM, RPS6KA3, FOXA1, EGFR, and TP53** contribute to the **positive regulation of transcription by RNA polymerase II** (GO:0045944) and **DNA-templated transcription** (GO:0045893). Notably, **TP53, EGFR, MET, and ATM** are linked to **upper respiratory tract disease**, suggesting that pre-existing lung conditions may contribute to cancer development. The driver genes **TP53, EGFR, ERBB2, MET, PPP2R1A, RPS6KA3, GNAS, and ATM** are involved in key pathways, including **central carbon metabolism, MAPK signaling, HPV infection, and microRNA-mediated regulation**. Reactome analysis further highlights their roles in **growth factor transcription regulation, PIP3-AKT signaling activation, TP53 degradation control, and caspase modulation**, all of which are critical for tumor survival, proliferation, and evasion of apoptosis.



Enrichment network of driver genes in male lung adenocarcinoma

The enrichment network for **male LUAD** shows that **TP53** and **EGFR** regulate the **positive production of miRNAs** involved in **gene silencing** (GO:1903800), while **EGFR**, **PPP2R1A**, **ERBB2**, **RPS6KA3**, and **TP53** participate in **cell growth regulation** (GO:0001558) and **positive growth signaling** (GO:0045927). **MET**, **ATM**, **RPS6KA3**, **FOXA1**, **EGFR**, and **TP53** contribute to **transcriptional regulation** by **RNA polymerase II** (GO:0045944) and **DNA-templated transcription** (GO:0045893). Furthermore, **KEGG analysis** connects the driver genes to **MAPK signaling**, **central carbon metabolism**, **HPV infection**, and **microRNA pathways**, demonstrating their involvement in **growth**, **survival**, and **metabolic adaptation**. **Reactome pathways** further reveal regulation of **growth factor transcription**, **PIP3-AKT signaling**, **TP53 degradation**, and **caspase activation**, while **ChEA3 transcription factor analysis** (**E2F1**, **BACH1**, **ISL1**, **RNF2**, **SUZ12**) indicates coordinated **upstream control**.

TRANSLATIONAL INSIGHTS: THERAPEUTIC TARGETS AND DRUG REPURPOSING

Based on the disrupted mechanisms in each cancer subtype and the associated mutated genes, drugs were evaluated for repurposing. A higher interaction score indicates a stronger predicted efficacy or better target match. The status indicates whether the drug is officially approved for that indication, while some drugs may be used off-label.

Gene	Drug	Indication	Mechanism	Status	Interaction score
BRAF	Dabrafenib	NSCLC	BRAF inhibitor	Phase II	2.47
TP53	Eprenetapopt	Ovarian	P53 stabilizer	/	1.03
TP53	Ipilimumab + Nivolumab	NSCLC	CTLA-4+PD-1 checkpoint blockade	Clinical study	0.01

Drug–gene interactions for female LUSC, showing genes, matched drugs, approval status, and interaction scores. Higher scores indicate stronger predicted targeting

Gene	Drug	Indication	Mechanism	Status	Interaction score
KEAP1	Dimethyl fumarate	/	Kelch-like ECH-associated protein one inhibitor	/	4.74
TF	Transferring-receptor-targeted anti-RRM2 siRNA	/	siRNA via transferrin receptor	/	6.52
EGFR	Afatinib	Lung cancer	Tyrosine kinase inhibitor (EGFR, HER2)	Approved	1.22
HDAC1	Vorinostat	Melanoma, colorectal cancer	HDAC inhibitor	Preclinical – cell culture	1.53

Drug–gene interactions for male LUSC, showing genes, matched drugs, approval status, and interaction scores. Higher scores indicate stronger predicted targeting

Gene	Drug	Indication	Mechanism	Status	Interaction score
EGFR	Afatinib	Lung cancer	Tyrosine kinase inhibitor (EGFR, HER2)	Approved	1.22
ERBB2	Pyrotinib	/	Receptor protein-tyrosine kinase erbB-2 inhibitor	/	0.17
TP53	Eprenetapopt	Ovarian	p53 stabilizer	/	1.03
PPP2R1A	Entrectinib	/	TRK / ROS1 / ALK inhibitor	/	0.02
MET	Tivantinib	Lung cancer	MET inhibitor	Not approved	3.51
ATM	Durvalumab + Olaparib	Neuroendocrine carcinoma	Immune checkpoint + PARP inhibition	Case Reports/Case Series	0.33
BCOR	Methotrexate	/	DHFR inhibitor	/	0.47
GNAS	Amitriptyline hydrochloride	/	Tricyclic antidepressant; potential off-target effect	/	0.03

Drug–gene interactions for female LUAD, showing genes, matched drugs, approval status, and interaction scores. Higher scores indicate stronger predicted targeting

Gene	Drug	Indication	Mechanism	Status	Interaction score
KRAS	Panitumumab	Colorectal	EGFR-targeted therapy (KRAS activation)	FDA rejected	0.97
KEAP1	Dimethyl fumarate	/	KEAP1 inhibitor	/	4.74
SMAD4	Gemcitabine	/	Nucleoside analog; inhibits DNA synthesis.	/	0.11
STK11	Everolimus	/	mTOR inhibitor	/	0.46
EGFR	Afatinib	Lung cancer	Tyrosine kinase inhibitor (EGFR, HER2)	Approved	1.22

Drug–gene interactions for male LUAD, showing genes, matched drugs, approval status, and interaction scores. Higher scores indicate stronger predicted targeting

FUTURE DIRECTIONS

Larger, multi-ethnic cohorts of non-smoker NSCLC patients are needed, with integration of **genomic, transcriptomic, epigenomic, and metabolomic** data. **AI/Machine Learning approaches** can improve subtype classification, survival prediction, and drug response modeling. Sex-specific analyses, such as hormonal regulation and **drug repurposing pipelines** linked to omics findings should be prioritized.

CONCLUSION

NSCLC subtypes remain poorly understood in non-smokers. Limited gender-specific and multi-omic studies hinder progress. **Systems-level, multi-omics research** is essential for advancing **biomarker discovery** and **personalized therapies** to improve patient outcomes.

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